

Supporting Information

Probabilistic dengue and chikungunya forecasting in Brazil: metric-dependent model rankings, regime-shift fragility, and conformal ensembles

This document provides the scoring definitions (S1), full model configurations (S2), the conformal recalibration and ensemble-composition robustness analysis (S3), normalized WIS by season (S4), the EPIFORGE 2020 reporting checklist (S5), exploratory data analysis (S6), and the reproducibility statement (S7). All numbers correspond to the committed backtest and are reproducible from the released pipeline.

S1 Text. Scoring

Forecasts are submitted as the nine quantiles $\tau \in \{0.025, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95, 0.975\}$ per state and week, which define the median and the four central prediction intervals at nominal levels 50, 80, 90, and 95 percent. The weighted interval score is

$$\text{WIS}(F, y) = \frac{1}{K + \frac{1}{2}} \left[\frac{1}{2} |y - m| + \sum_{k=1}^K \frac{\alpha_k}{2} \text{IS}_{\alpha_k}(\ell_k, u_k, y) \right], \quad \text{IS}_{\alpha}(\ell, u, y) = (u - \ell) + \frac{2}{\alpha}(\ell - y)\mathbf{1}\{y < \ell\} + \frac{2}{\alpha}(y - u)\mathbf{1}\{y > u\}$$

with m the predictive median, $K = 4$ central intervals, and $\alpha_k = 1 - L_k/100$. This equals the mean pinball loss over the nine quantiles up to the constant weighting, the exact discrete approximation to the continuous ranked probability score. We verified our implementation against the challenge platform’s own scorer to machine precision on a shared set of forecasts, and against a hand-computed toy case that caught an early normalization error (division by 9 rather than $K + \frac{1}{2} = 4.5$). Alongside mean WIS we report: empirical interval coverage; a decomposition of WIS into a dispersion term and directional over- and under-prediction penalties; a per-unit relative WIS that scores each state-season-horizon against the naive baseline and takes a geometric mean, which weights every unit equally; and the normalized WIS, the sum of WIS divided by the sum of observed cases, which is the challenge’s headline metric and is scale-free across regions.

S2 Text. Model configurations

Features (gradient boosting). The LightGBM model is fit on a synthetic-origin panel that expands each fold into many weekly origins crossed with forecast horizons 1 to 67 weeks, with origin-anchored features computed only from the cutoff-filtered history:

- *Autoregressive:* log-incidence lags at 1, 2, 3, 4, 8, and 52 weeks; rolling mean over 4 and 8 weeks; rolling standard deviation over 4 weeks; rolling maximum over 12 weeks.

Table 1: S1 Table. Frozen model configurations.

Model	Configuration
LightGBM (per quantile)	objective: quantile at each of the nine levels; num_leaves 31; max_depth 5; learning_rate 0.05; 400 boosting rounds; feature_fraction 0.8; bagging_fraction 0.8; bagging_freq 1; seed 42; deterministic. Intervals calibrated by conformalized quantile regression on a trailing 78-week window.
GRU deep ensemble	input sequence length 104 weeks; single-layer GRU encoder, hidden size 48, shared across all 26 states; per-state embedding of dimension 8; decoder is a two-layer perceptron with hidden size 64, ReLU, and dropout 0.1; negative-binomial output head; quantiles analytic from the predictive distribution; deep ensemble of 5 independently seeded runs pooled by quantile averaging; 40 epochs, Adam, learning rate 10^{-3} .
Semi-mechanistic	800 bootstrap resamples of whole historical season trajectories; season length 53 weeks; negative-binomial observation model; a local reimplementa-tion of Richards-curve epidemic-scanner fits.
Ensembles	per-quantile median (Vincentization) of the climatological, gradient-boosted, and deep-learning models; and inverse-WIS weighting tuned only on the designated tuning season.

- *Seasonality*: target epidemiological week; first and second sine and cosine harmonics; a target-relative seasonal anchor equal to the most recent same-target-epiweek incidence at or before the origin.
- *Observed climate (four-week means at the origin)*: temperature, precipitation, and relative humidity, plus a temperature anomaly against each state’s same-epiweek climatology.
- *Seasonal climate forecast*: the seasonal forecast of temperature, humidity, and precipitation for the target month, merged leakage-safely so that only forecasts issued at or before the origin are used.
- *Ocean oscillation indices*: El Niño–Southern Oscillation, Indian Ocean Dipole, and Pacific Decadal Oscillation, each at lags 0, 4, 8, 12, 26, and 52 weeks.
- *Static*: log population; population-weighted Köppen climate-zone fractions; biome fractions; forecast horizon in weeks.

Hyperparameters (S1 Table). Table 1 lists the frozen configuration of each model. These were chosen a priori as conservative defaults. A separate search that preferred a deeper boosted configuration (num_leaves 63, max_depth 7) improved an internal time-block holdout by about 5 percent but worsened the true backtest, and was not adopted (main text the main text, negative results).

S3 Text. Conformal recalibration and ensemble composition

Conformal recalibration. For each central interval level L with bounds $[\ell_L, u_L]$ and median m , and for each calibration unit, the multiplicative factor needed to just cover the observation is $r = (y - m)/(u_L - m)$ when $y > m$ and $r = (m - y)/(m - \ell_L)$ when $y < m$, and 0 otherwise. The widening factor s_L is the $L/100$ empirical quantile of r over the tuning season, so that widening the

Table 2: S2 Table. Ensemble composition sweep (each subset with conformal recalibration). nWIS is normalized WIS. The chosen composition is in bold.

Members	Mean WIS	nWIS all	nWIS ordinary
Climatological + Mechanistic	1185	0.541	0.420
Climatological + GRU + Mechanistic	1213	0.554	0.387
Climatological + LightGBM + GRU (chosen)	1216	0.555	0.333
Climatological + GRU	1216	0.555	0.326
GRU + Mechanistic	1217	0.556	0.339
Climatological + LightGBM + GRU + Mechanistic	1247	0.569	0.356
LightGBM + GRU	1299	0.593	0.331
LightGBM + Mechanistic	1314	0.600	0.408

interval about the median by s_L yields approximately L percent empirical coverage. The estimated factors are $s_{50} = 1.03$, $s_{80} = 1.00$, $s_{90} = 1.23$, and $s_{95} = 1.73$. They are gentle and concentrate on the outer tails, which is why the recalibration protects against catastrophic under-prediction in the outbreak season at small cost in ordinary seasons. Recalibrating the ensemble output outperforms recalibrating any single member, because the per-quantile median across members otherwise dilutes a member’s widening (recalibrating the deep-learning member alone yields mean WIS 1259 versus 1216 for the ensemble-output recalibration).

Member-composition robustness (S2 Table). Table 2 scores every member subset under the same conformal recalibration. The chosen three-member ensemble is close to the best composition by the headline normalized WIS, and no subset is better on both the all-season and the ordinary-season metric. The subset that appears best by all-season mean WIS (climatological plus mechanistic) is the worst of the strong subsets on ordinary seasons (normalized WIS 0.420 versus 0.333), because it wins only by handling the 2024 outlier and drops the two machine-learning models that carry the ordinary-season signal. This is the composition-level analogue of the metric-disagreement finding.

Weight robustness. Inverse-WIS ensemble weights are stable across the choice of tuning season: tuning on fold 1 gives (climatological 0.31, LightGBM 0.34, GRU 0.35); on fold 2, (0.33, 0.36, 0.30); on fold 3, (0.31, 0.25, 0.45). The unweighted median ensemble avoids this choice entirely and performs comparably, which is why it is preferred.

S4 Table. Normalized WIS by season

Table 3 gives normalized WIS by model and season. It shows the regime split numerically: the deep network is best on the ordinary seasons (fold 3, 0.271) but worst on the 2024 outlier (fold 2, 0.798), while the conformal ensemble has the best outbreak-season normalized WIS of any ensemble (0.671) and is therefore best overall.

S5 Text. EPIFORGE 2020 reporting checklist

Table 4 maps the recommended reporting items for epidemic forecasting research to their location in the manuscript and this Supporting Information.

Table 3: S4 Table. Dengue normalized WIS by season. Lower is better. Best in each column in bold.

Model	Fold 1	Fold 2	Fold 3	Fold 4
Ensemble (conformal)	0.308	0.671	0.335	0.477
Ensemble (Vincentization)	0.308	0.720	0.325	0.436
Ensemble (inverse-WIS)	0.292	0.727	0.321	0.507
LightGBM	0.320	0.671	0.492	0.866
Mechanistic	0.425	0.680	0.421	0.552
Climatological	0.350	0.724	0.394	0.446
GRU	0.308	0.798	0.271	0.451
Seasonal-naive	0.454	0.776	0.444	0.528
Naive	0.659	0.773	0.806	0.653

Table 4: S3 Table. EPIFORGE 2020 checklist mapping.

Item	Location
Study purpose and forecasting objectives	Introduction; Results, study design
Forecast targets and horizons	Results, study design; Methods, data and targets
Data sources and provenance	Methods, data and targets; S2 Text
Data pre-processing and covariates	Methods; S2 Text (feature list)
Model descriptions and assumptions	Methods, models; S1 Table
Uncertainty quantification	Methods, scoring and conformal; S1, S3 Text
Validation approach and leakage control	Methods, backtest design
Performance metrics	Methods, scoring; S1 Text
Baseline and comparator models	Results; Tables in main text
Results with uncertainty and calibration	Results; Figures 2 to 5
Interpretation, generalizability, limitations	Discussion
Data and code availability	Data and code availability; S7 Text

S6 Figs. Exploratory data analysis

The following figures summarize the data that motivated the modeling choices. S1 Fig shows the per-state seasonal climatology of dengue incidence. S2 Fig shows the lagged cross-correlation between climate variables and incidence. S3 Fig shows incidence stratified by Köppen climate zone. S4 Fig relates annual dengue totals to the El Niño–Southern Oscillation. S5 Fig shows the distribution of epidemic peak weeks across states and seasons.

S7 Text. Reproducibility and provenance

The complete pipeline is released as an installable package with a single command to reproduce every result. Leakage safety, scoring correctness, model persistence, and every model family are covered by an automated test suite that runs in continuous integration. All model fitting is deterministic under a fixed seed: gradient boosting uses LightGBM deterministic mode, and the deep ensemble fixes the seed of each member. A provenance manifest records the exact data checksums and code revision behind each reported number, and the per-state scored predictions for all models are released so that any table or figure can be regenerated. Validated submission files for the four official tracks (dengue and chikungunya, at state and city level) are included.

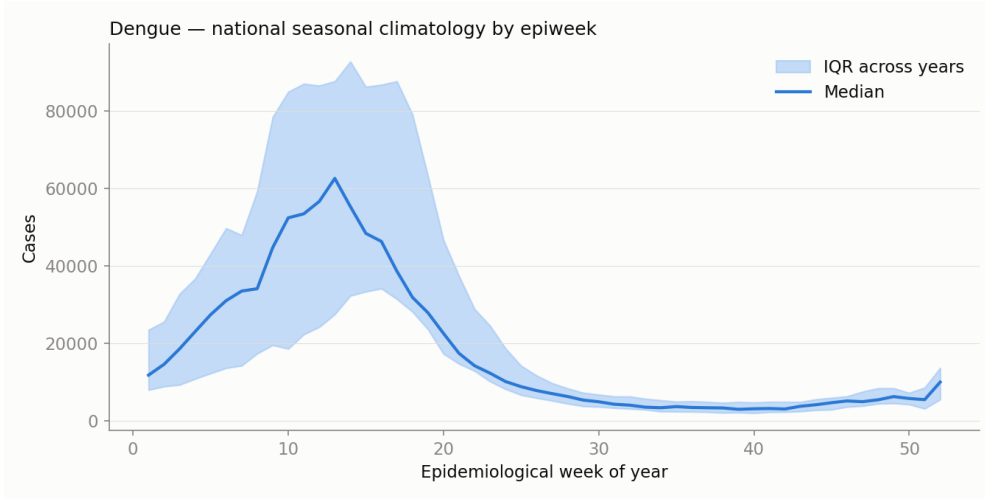


Figure 1: S1 Fig. Seasonal climatology of dengue incidence by state.

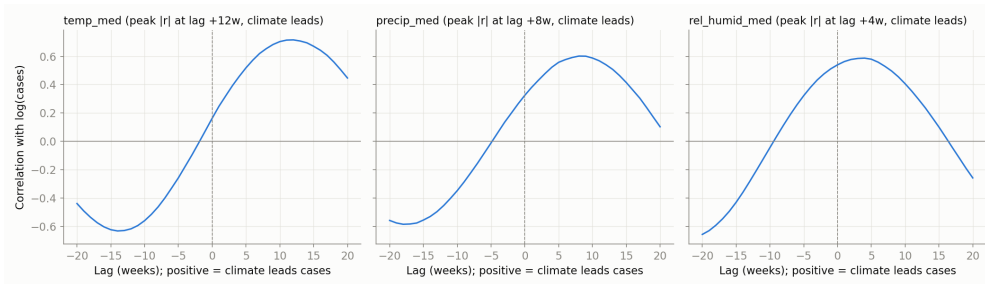


Figure 2: S2 Fig. Lagged cross-correlation of climate variables with dengue incidence.

S8 Text. Per-state model performance heatmap

Table-based per-state comparisons do not scale to six models across 26 states. Fig S6 gives the same comparison as a heatmap: each cell is one model’s mean log relative WIS against the seasonal-naive baseline in one state, averaged over the four seasons, the same quantity underlying the stability figure in the main text (Fig 3) but broken out by state instead of collapsed to a mean and standard deviation. Green cells are better than the baseline and red cells are worse. Two patterns are visible beyond the aggregate numbers already reported: the deep network’s advantage is concentrated in a subset of states (Ceara and Rio de Janeiro most strongly) rather than uniform, and the gradient-boosted model is the only one of the six with states that are red, meaning worse than the seasonal-naive baseline (Amapa, Paraiba, Pernambuco), which the state-level aggregate in Table 1 does not show.

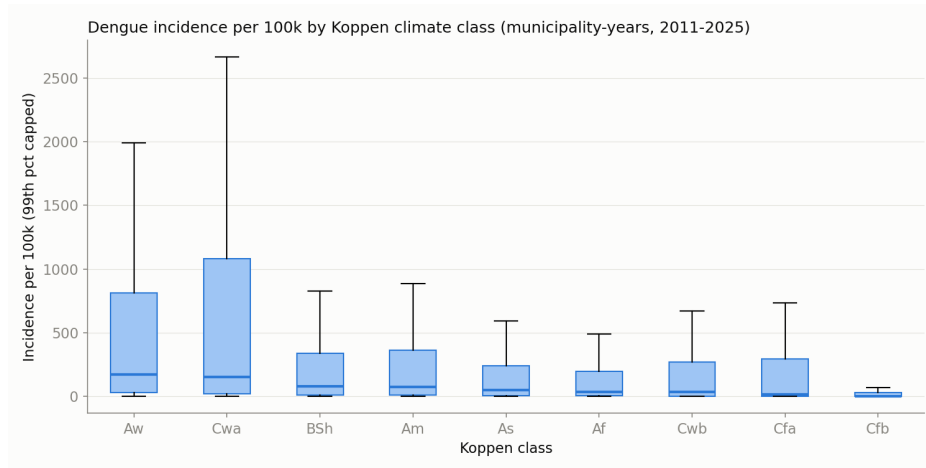


Figure 3: S3 Fig. Dengue incidence by Köppen climate zone.

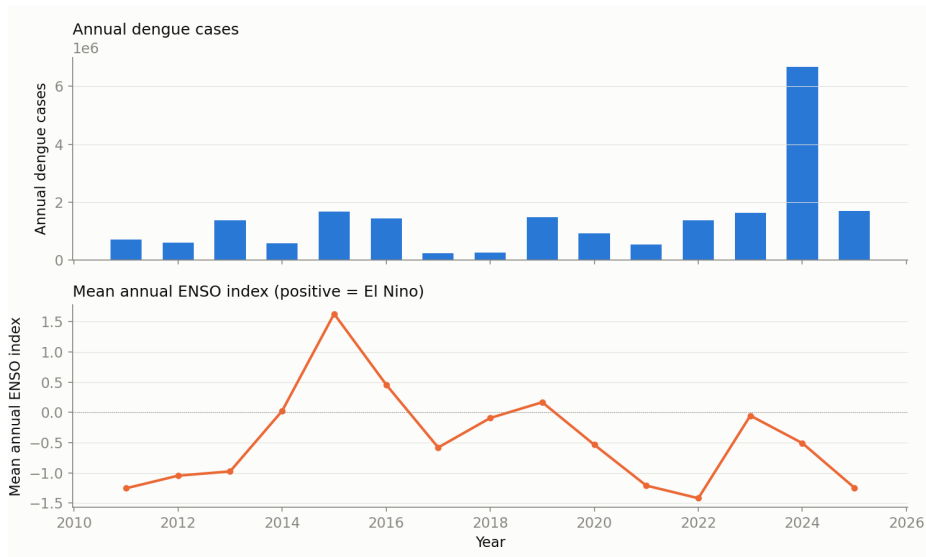


Figure 4: S4 Fig. Annual dengue totals versus the El Niño–Southern Oscillation index.

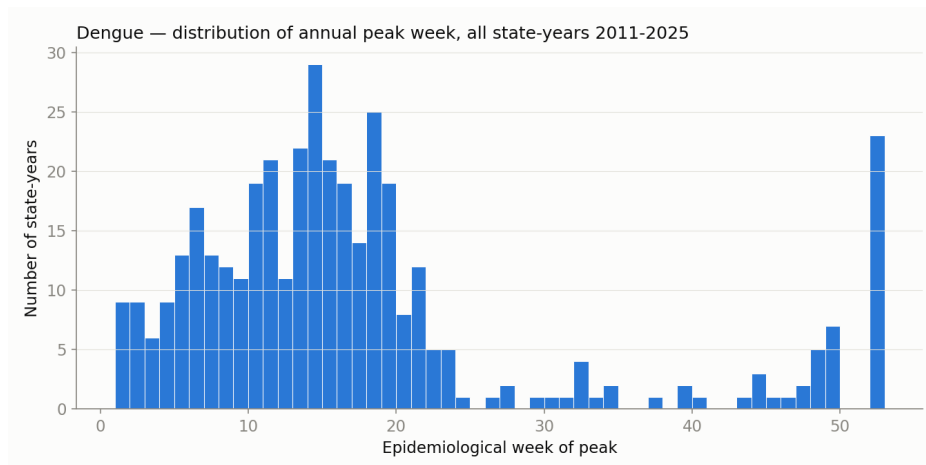


Figure 5: S5 Fig. Distribution of epidemic peak weeks across states and seasons.

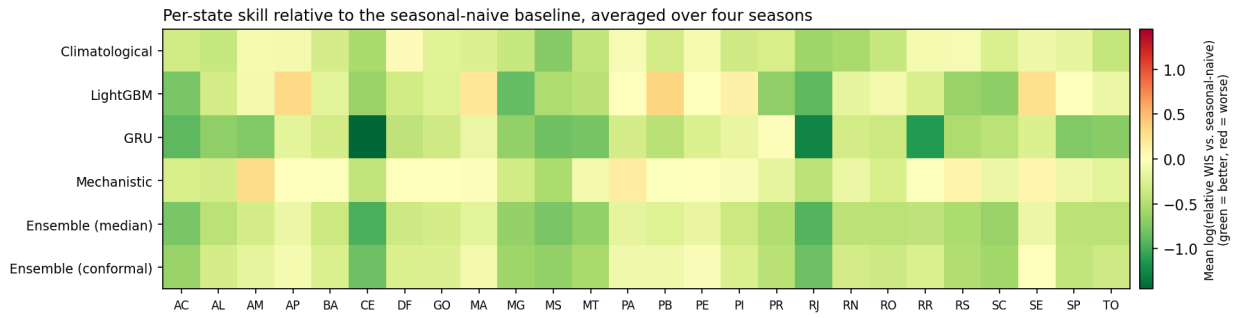


Figure 6: S6 Fig. Per-state skill relative to the seasonal-naïve baseline. Mean log relative WIS by model and state, averaged over the four seasons. Green is better than baseline, red is worse.